

Date	07/07/26
Team ID	XXXXXX
Project Title	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

PROJECT INITIALIZATION AND PLANNING PHASE

Project Proposal (Proposed Solution) report

The proposal report aims to transform personalized education and student support using the state-of-the-art Google Gemini machine learning models, significantly boosting learning efficiency and structural accuracy.

It tackles modern remote educational inefficiencies, promising enhanced accessibility, comprehensive automated study material generation, and an engaging interactive experience for learners.

Key features include an advanced LLM-based cognitive tutoring model and real-time adaptive decision-making tailored to each student's specific academic requirements.

Project Overview	
Objective	The primary objective is to revolutionize the personalized learning process by implementing advanced generative AI techniques via Google Gemini, ensuring faster, more personalized, and highly accurate student assessments.
Scope	The project comprehensively assesses and enhances the digital self-learning framework, incorporating generative models for a more robust, conversational, and effective automated support system.
Problem Statement	
Description	Addressing structural gaps, rigid static curricula, and lack of personalized interactions in current e-learning platforms, which adversely affects student engagement, completion rates, and comprehensive academic satisfaction.
Impact	Solving these issues will result in vastly improved learner understanding, personalized progression speeds, and an overall optimization of the educational landscape, directly contributing to career readiness and institutional success.
Proposed Solution	
Approach	

	Employing Google Gemini large language models to analyze student queries and performance metrics, creating a dynamic, safe, and highly adaptable intelligent learning assistant.
Key Features	• Implementation of an LLM-based adaptive credit/knowledge assessment framework.

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Cloud Server instances / API processing allocations	Google Vertex AI API / T4 GPU
Memory	System RAM for hosting interface application	16 GB
Storage	Disk space for application files, user cache, and log tracking	500 GB SSD
Software		
Frameworks	Python backend framework	Flask / FastAPI
Libraries	Core development and API interface libraries	google-generativeai, langchain, pandas, numpy, matplotlib
Development Env	Integrated Development Environment	Jupyter Notebook, VS Code
Data		
Data (Kaggle)	Source educational datasets, question-answering pairs	Kaggle Open Dataset, CSV/JSON
Data (Custom)	Academic curriculum guidelines and structural indices	Textbook Corpus, PDF format

SKILL WALLET

A SmartBridge Product

- Real-time decision-making for quicker academic guidance and contextual feedback.
- Continuous feedback loops to adapt dynamically to evolving individual student learning speeds and landscapes.